CAD can start. Fabrication stays on hold.

Evidence checked 15 Oct 2026 UTC / 14 Oct Toronto

This package freezes the owner-approved control inventory and supplies one proposed coordinate schedule. It does not contain a native enclosure model, STEP export, manufacturing drawing or physical test pass.

Decision	Current result
Preliminary CAD	YES. Build the proposed layout in Onshape and label uncertain features HOLD.
Preliminary DFM submission	Conditional. First create the actual model, STEP and dimensioned PDF; this audit supplies the design basis.
Enclosure fabrication	NO. Exact key/joystick/display stacks, rear parts, supports, clearances, owner layout and Protocase acceptance remain open.
Commissioning	NO. Native electronics/firmware, one-head and four-link tests remain outstanding.

What is fixed

34 keys, six encoders, 27 independent illuminated keys. Ten CAM keys, twelve selected-camera presets, twelve technical keys. ServoCity605616 is retained. APEM1KC concave/frosted-white/red appearance is retained. AUTO FOCUS is below the focus knob. The owned Nextion must work without touch.

Most consequential corrections

- Use the 17.75 mm APEM bezel envelope, not only the 14.3 mm cap. Its 14.7 mm minimum uncoated opening is not a released powder-coated cut.

- The Nextion window is offset within its PCB. Confirm the actual module; manufacturer active-width references conflict.

- Six input registers still fit46 inputs. LED outputs increase to fourHC595 plus27 sink stages; the old12-LED circuit is obsolete.

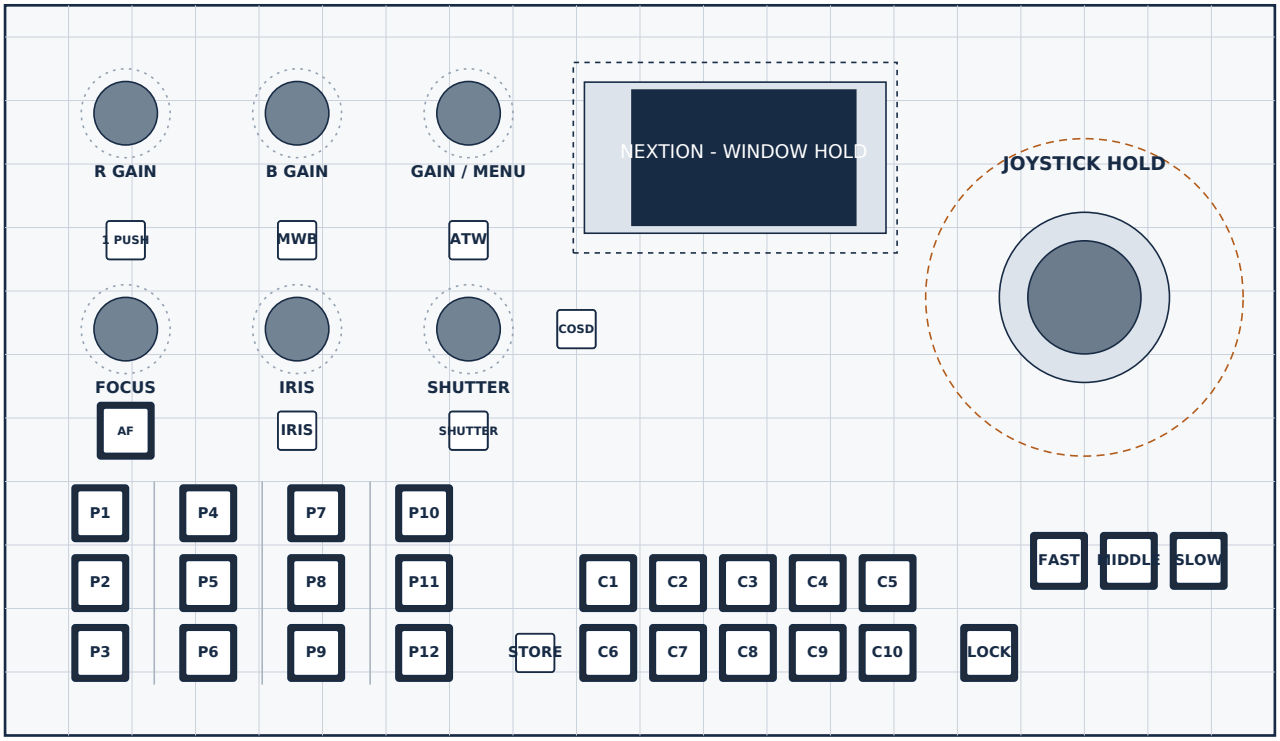
- C$500 Protocase support includes shipping. November6 delivery and taxes are unresolved. APEM assistance and W55 shipment remain pending.

One layout, all controls

All coordinates are proposed. Full control schedule: dimensional workbook.

400 mm proposed width; 230 mm along slope

REAR / uphill



FRONT / operator - origin (0,0)

Parameter	Proposed / calculated value
Sloped panel	400 x230 mm at10 degrees
Body plan / heights	400 x226.506 mm; front55, rear94.939 mm
Material / construction	Stock5052-H32,1.6256 mm candidate; removable top/rear/bottom
Packing basis	27 bezels17.75 square;7 caps12.15;6 knob operating circles28; display102x60; joystick trial hand circle100

The2D envelope calculation found no overlap. It did not test hand comfort, labels, standoff levels, moving joystick geometry, cable paths, corner joints or panel deflection. The100 mm hand circle and board/cable reservations are project allowances. Confirm the proposed six encoder functions and IRIS/SHUTTER mode legends before engraving.

Panel datum: front-left, Xright, slopeYrear/uphill. World point(X,s,n)=(X,s cos(a)-n sin(a),Hf+s sin(a)+n cos(a)). Rear drawings mirrorX. Do not mix PCB, panel-normal and world-height dimensions.

The mechanical release blockers

Manufacturer dimensions are references; actual fit remains untested.

Part	Reference and unresolved fit
APEM 1KC /2K [1]	Complete candidate3FTH9871KC11162K09. Cap14.3; bezel17.75; lower bezel16.4. Drawing19.1 concave/20.1 rim references are not panel protrusion. Exact footprint/support datum, retention and coupon required.
605616 [2]	Flange diameter53.594; total85.09; four2.54 holes; Ypitch32.512. Xpitch, actual opening, second body axis, solder tabs and swept/hand volume remain HOLD. M2 candidate; M3 does not fit nominal part holes.
Nextion [3]	PCB95x47.6; holes3.2 on88.6x41.2. Official active widths69.60/69.90 conflict. Trial70.5x42.6 window is offsetX+2.73 fromPCB centre. Verify owned suffix, bezel and mated cable.
B3F/B32 [4]	Seven of25 owned bare switches used;18 remain. Cap12 square; assembled top10 abovePCB. Uncoated aperture12.20;12.8 coupon trial only. Independent support plane may be needed.
Encoder/knob [5]	PEC11R-4220F-N0024 has no push switch. M7x0.75,6 mm shaft. Supplied nut/washer included. Knob5527 diameter20; page12.5 high vsdrawing15.5 overall. Verify engagement and soldering method.
W55 [6]	PCB80x21, full model85.35x21x16.6. CAD-derived holes(78,3.5)/(78,17.5), diameter2.0. Actual board revision, sockets, plugs and rear alignment remain open.
Rear I/O [7]	Four RJHSE5080 serial-jack uncoated reference openings16.129x13.208,R0.254. Ethernet is a different jack. DC/USB exact parts, PCB brackets and finished apertures remain HOLD.

Finish and forming [8]

Candidate stock radius1.5748 and15 mm flanges need supplier DFM. A current tooling row lists7.62 mm minimum flange and5.08 mm small-hole distance from outside bend corner; arbitrary cutouts use the flange restriction. Do not apply that small-hole rule to every opening. Current coating guidance is0.003 in per wall; exact quoted finish and process tolerances control production.

Calculate raw opening from the required finished opening, two-wall coating, negative cutting tolerance and registration budget. Do not add the same component tolerance twice. A numerical clearance budget does not replace a finished-panel coupon.

Electrical and camera scope

Static architecture fits on paper; hardware operation is not validated.

Resource	Current basis
GPIO / UART	19 external signals. UART0GP0/1 CAM1; UART1GP4/5 display; GP6-11 cameraPIO; ADC26-28 joystick. GP2/3/12/13/14/15 scan/control.
Inputs / LED outputs	34+12=46 scanned inputs,6x8=48 bits.27 LEDs,4x8=32 output bits plus27 low-side sinks. SharedGP15 global disable cannot provide routine brightness PWM.
PIO / DMA	Prior static SDK review includes1 EthernetSM+6 cameraSMs;1 spare. DMA5 withcameraRX or8 withoptionalTX. No native firmware build or simultaneous-load pass.
Timing	48 clocks/250 kHz=192 us. A2 kHz frame has500 us:38.4% serial wire time, excluding overhead. Final32bits carryLED image; do not latch intermediate scan data.
Power / protection	Illustrative5V load1.096A;3A adapter candidate. VSYS4.3-5.5 after losses. USB arbitration, supply injection, inrush, fuse, AUX current/thermal and bus protection remain HOLD.

RS-422 and STOP [9]

AM26LV31E/32E remain conditional quad driver/receiver candidates. Open-circuit fail-safe is not terminated-disconnected fail-safe. Resolve actual cable pairs, reference/ground potential, termination, bias and entry protection. Serial contacts:1GND;2NC;3TX-;4RX-;5RX+;6TX+;7/8NC. Confirm actual jack pad view and cable continuity.

STOP must cover pan/tilt, zoom and focus. LOCK and CAM switch need stop/neutral re-arm. Power/cable loss can prevent STOP from reaching a moving head; remote watchdog behaviour is unverified. ExistingRMC fallback must not drive the same pair in parallel.

Presets and Panasonic functions [10]

PTR-10T lists50 pan/tilt positions. SuppliedV2.7E4 generic memory command accepts pp0-127; proposedP1-P12 IDs0-11 require head/firmware qualification in agreed spare slots. ACK means accepted, completion means executed; actual arrival semantics must be observed/qualified before showingREACHED. A timer is insufficient.

AG-CX350PX zoom/focus is listed in the supplied non-T manual. ActualT model, Panasonic variant/cable and command path still need matching. Iris/shutter/gain/WB/ATW/cameraOSD andAFmode feedback remain unavailable until verified. Retain controls physically without inventing support.

Purchasing and Friday handoff

Friday is a DFM target, not automatic fabrication authorization.

Current priced subset	Qty	CAD before tax
B32-1310 caps	7	4.13
PEC11R encoders	6	21.06
Adafruit5527 knobs	6	28.26
ServoCity605616	1	45.70
GST18U05-P1J supply	1	21.12
Subset total		120.27
Subset with13%HST		135.91

The subtotal excludes27 complete APEM assemblies, unpriced electronics, AC cord, PCB fabrication, freight and enclosure overage. The full remaining cost isUNKNOWN. Start paper CAD forC$0; optional mechanical sample basketC$55.09, orC$76.21 including the later supply, before tax/freight. No purchase was placed.

Protocase confirmedC$500 manufacturing support covering finish, graphics, hardware and shipping. Taxes are unresolved. APEM already received the specific sample/drawing/quantity27 request; no reply found. W55 sample approval is not shipment or receipt.

Close eight gate families

G1 APEM complete assembly: Quote/MOQ/lead time; assembly drawing/footprint and PCB-to-panel stack; finished coupon.

G2 605616 joystick: Measure both hole pitches and actual opening; moving and hand sweep; tabs and wire exit.

G3 Owned Nextion: Rear-label suffix/revision; window offsets; hole pattern and mated cable depth.

G4 Seven B3F/B32 + six encoder/knobs: Coated coupon, cap travel, knob engagement and all standoff heights; soldering method.

G5 Full layout and functions: Approve42 control centres (34keys+6enc+display+joystick), proposed 400x230 surface and labels; six encoder/mode mapping.

G6 Rear I/O and power: Exact LAN/DC/USB openings, connector orientation, fastening, PCB brackets and cable withdrawal.

G7 Board supports and interference: Native PCBs or bracket design; mounting grid; full3D interference and lift-out/strain relief test.

G8 Sheet-metal DFM and quote: Stock material/radius/fasteners; seam/finish tolerances, joystick load path, quote and owner approval.

Submit the actual editable CAD reference, assembly STEP inmm and dimensioned PDF, with matchingrevision, quantity1, material/finish/legends and visible HOLDs. Do not email this analysis as if it were an enclosure CAD file. Owner approves production after Protocase review and quote.

What was actually checked

Source access, verification and remaining actions

Recovered and read: attached brief; visible project decisions; PR2/R6.3-R6.7; R6.2 machine-readable BOM, mechanical/electrical notes, pinmap and protocol review; historicalKiCad check JSONs; primary component drawings and current supplier communications. Full earlier conversation exports, originalBOMv3, actualOnshape model and revised nativeKiCad were unavailable. Separate earlier source files were not all reread; inherited lineage is labelled.

Verification run: all42 control envelopes checked numerically in2D;34keys and27LEDs reconciled; all96 legacyBOM IDs mapped. Workbooks recalculated, changed-quantity/quote/slope/bounds cases restored, formula errors scanned, and every worksheet previewed. NativeExcel/LibreOffice was not run. ExistingPython assertion script and four unittest cases passed this session. These are host simulations, not embedded or camera tests. HistoricalKiCad still reports3 unconnected nets and7 powerERC errors; no newKiCad pass is claimed.

Still required: actual Onshape model/exports, owned display and complete-control measurements, finished coupons, native3D interference/assembly review, final power/protection/PCB design, one-head tests, qualified preset/arrival/STOP feedback and four-link stress tests. To revise an existing CAD model, provide its actualOnshape document or current source/export; an image is insufficient.

Primary source links

1 APEM assembly family
https://www.apem.com/storage/datasheets/files/MEC_switches_serie_1K_1KB_1KC/media/MEC-switches-serie-1K-1KB-1KC.pdf

2 ServoCity mechanical gallery
https://www.servocity.com/3-function-joystick/

3 Nextion family and linkeddrawing
https://nextion.tech/datasheets/nx4024t032/

4 Omron/Aratas B32 drawing
https://omronfs.omron.com/en_US/ecb/products/pdf/en-b32.pdf

5 Bourns encoder RevD04/26
https://www.bourns.com/docs/product-datasheets/pec11r.pdf

6 W55 hardware and authenticSTEP
https://docs.wiznet.io/Product/Chip/MCU/W55RP20/w55rp20-evb-pico

7 Amphenol jack RevJ
https://cdn.amphenol-cs.com/media/wysiwyg/files/drawing/p-rjhse-5080.pdf

8 Protocase forming/finish
https://www.protocase.com/resources/bend-radii/

9 TI line driver/receiver
https://www.ti.com/lit/ds/symlink/am26lv32e.pdf

10 Datavideo protocol /T manual
https://datavideo.com/ap/protocol/181/ptr-10-mark-ii-visca-control-protocol

11 RMC E4 reference manual
https://www.bhphotovideo.com/lit_files/846207.pdf

12 Panasonic remote-control instructions
https://pro-av.panasonic.net/manual/html/AG-CX350P.PJ.EN.ED.AN.PX(DVQP1830ZA)_E/chapter08_01_02.htm

13 Tax rate
https://www.canada.ca/en/revenue-agency/services/tax/businesses/topics/gst-hst-businesses/charge-collect-which-rate/calculator.html

Exact component/purchasing links and source classifications are also preserved beside their workbook rows. The T manual is manufacturer-authored on a supplier mirror; latest E7/E8 full manuals were not asserted reviewed. Private shipping addresses, phone numbers and billing attachments are excluded.